



Regulatory Complaint Investigation and CAPA Report

Confidential - AI Draft

Document No.	DP-M4-20260708-THELOWBA
Classification	Confidential - AI Draft
Generated	2026-07-08T10:40:47Z
Device	DocPlus+ Pulse Oximeter (DEV-PULSE-OX)
Firmware	v3.4
Framework	ISO 13485
Prepared By	DocPlus+ M4 Evidence and Documentation Layer

This document was generated from live M1 graph facts and M2 agent outputs. Every summarized claim is tied to captured agent state, evidence citations, or audit findings.

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This output follows a formal regulatory-document pattern: controlled metadata first, then complaint facts, investigation logic, objective evidence, risk decision, CAPA plan, review gates, and approval placeholders.

1. Document Profile and Controlled Metadata

Document Title	DocPlus+ Regulatory Complaint Investigation and CAPA Report
Document Number	DP-M4-20260708-THELOWBA
Revision	A - AI Generated Draft
Complaint ID	CPL-20260708-53B0
CAPA ID	CAPA-20260708-6935
RTM ID	RTM-DP-53B06935
Classification	Confidential - AI Draft for quality and regulatory review
Generated Timestamp	2026-07-08T10:40:47Z
System of Record	DocPlus+ M4 Evidence and Documentation Layer
Device	DocPlus+ Pulse Oximeter (DEV-PULSE-OX)
Current Firmware	v3.4
Affected Component	COMP-POWER-BATTERY
Regulatory Framework	ISO 13485

Data Quality and Source Provenance

This report references 50 graph fact(s): 13 controlled, 22 extracted, 4 inferred, 9 synthetic, 2 internal.

Uncontrolled graph facts appear in this report and are visibly tagged. [DEMO DATA - NOT CONTROLLED EVIDENCE] means the item cannot support CAPA closure without controlled verification. [INFERRED DATA - REVIEW REQUIRED] means the item was inferred and requires reviewer confirmation before final use.

Source Type	Referenced Fact Count
controlled	13
extracted	22
inferred	4
synthetic	9
internal	2

Digital-Twin Prototype Disclaimer

Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Document Purpose

This report documents the complaint intake, AI-assisted investigation record, retrieved objective evidence, risk decision, CAPA recommendations, audit-style gaps, and corrective action controls. It is intended as a controlled draft that a Quality or Regulatory reviewer can verify, approve, or return for correction.

2. Regulatory Format Controls

ALCOA+ Data Integrity Mapping

Principle	DocPlus+ Implementation
Attributable	Prepared by DocPlus+ M4; generated timestamp and device scope are recorded.
Legible	A4 PDF with selectable text, structured headings, repeatable tables, and page numbers.
Contemporaneous	Generated at the time of complaint-report request from live agent state.
Original	Marked as AI Draft until Quality and Regulatory reviewers approve.
Accurate	Risk, CAPA, evidence, audit, and trace sections are populated from structured state.
Complete	Includes intake, investigation, evidence, risk, CAPA, audit, trace, and approval fields.
Consistent	Uses the same DocPlus+ controlled layout for all complaint-report exports.
Enduring	Saved as a PDF artifact under the application output directory.
Available	Downloadable through the M4 document endpoint and traceable by document number.

Applicable Regulatory Basis

Framework	Relevance to This Report
ISO 13485	Complaint handling, CAPA discipline, documented information, traceable quality records.
21 CFR Part 820	Design controls, complaint files, CAPA, and objective evidence expectations.
EU MDR	Technical documentation, post-market surveillance, vigilance, and risk-benefit continuity.
21 CFR Part 11	Electronic record controls, auditability, attribution, and durable retrieval.

3. Executive Summary

Severity Low	Risk Medium / RPN 10	Reportable False
Evidence Confidence medium (score 0.91)	Evidence Class Mix Controlled current verification: 10, Firmware traceability signal - informational, not controlled verification evidence: 2	Evidence Items 12
Audit Findings 3	Trace Decay Alerts 2	

Complaint: The low battery icon appears even though the device continues to operate for a long time.

Agent conclusion: Complaint: Low battery icon appears while the device continues operating for an extended period (severity: Low; affected component: COMP-POWER-BATTERY). Most plausible explanation aligns with prior closed complaint CPL-100-009, which found the "battery full" flag could be overwritten by a stale lowvoltage alarm persisted in nonvolatile settings. Controlled verification artifacts indicate related power/battery behaviors passed regression in firmware v3.4 (RUN-TC-PWR-009/002/003), so confirming the unit's firmware version and event logs is critical. An alternate possibility is user icon misinterpretation per CMP-006 (perfusion indicator vs battery icon).

Risk Evidence Confidence Basis

Weighted evidence-class score 0.91 from 12 item(s): Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=2. CAPA-blocking evidence-class item count: 2. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Confidence drivers: RUN-TC-PWR-009-V3-4: controlled_verification, RUN-TC-PWR-002-V3-4: controlled_verification, RUN-TC-PWR-003-V3-4: controlled_verification, RUN-TC-PWR-005-V3-4: controlled_verification, RUN-TC-PWR-007-V3-4: controlled_verification, RUN-TC-PWR-008-V3-4: controlled_verification, RUN-TC-PWR-010-V3-4: controlled_verification, RUN-TC-PWR-001-V3-4: controlled_ver...

Quality Review Decision

The complaint should remain open until objective evidence is reconciled, CAPA ownership is assigned, and any major or critical audit findings affecting the complaint scope are remediated or justified.

4. Complaint Investigation Record

Field	Generated Value	Regulatory Basis
Complaint ID	CPL-20260708-53B0	ISO 13485 8.2.2
Date Received	2026-07-08T10:40:47Z	21 CFR Part 11
Device UDI	UDI-DI-08717648-184649	EU MDR Article 27 / 21 CFR 830
Model / Version	DOCPLUS+-PULSE-OXIMETER / v3.4	IEC 62304 Section 8
Complaint Source	Post-market complaint intake	ISO 13485 8.2.2
Complaint Narrative	The low battery icon appears even though the device continues to operate for a long time.	21 CFR 820 complaint file
Patient Impact	Potential interruption or unavailability of device function.	EU MDR Article 87
MDR Reportable?	No	EU MDR Article 87 / 21 CFR 803
CAPA Initiated?	Yes - CAPA-20260708-6935	21 CFR 820.100
Affected Component	COMP-POWER-BATTERY	ISO 13485 8.5.2
Closure Status	Pending Engineering Approval - Simulated Evidence Sufficient: Formal Engineering review must approve the simulated/minor-gap findings and decide whether additional physical verification is required.	ISO 13485 8.2.2

Reportability Decision Logic

Condition	DocPlus+ Assessment	Action
Death or serious injury	Not stated in complaint text.	Monitor for escalation during intake and investigation.
Potential harm or near miss	Potential interruption or unavailability of device function.	Document rationale and continue risk review.
Device relationship	Complaint maps to Power and Battery, Connectivity and Data Transfer.	Keep CAPA open until engineering evidence confirms or rules out relationship.
Malfunction/use issue	Reported condition indicates possible device malfunction or use-related failure.	Verify recurrence risk and effectiveness of controls.

Related Prior Complaints

Complaint ID	Similarity	Summary	Source
CMP-006	4	User reported confusion between perfusion indicator and battery icon. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CPL-100-009	4	Low battery warning appears despite full charge. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CMP-003	2	Battery indicator dropped rapidly during extended monitoring. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CMP-012	2	MAUDE-style adverse event record linked to pulse oximeter use. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CPL-100-003	2	Battery drains unusually fast. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]

5. Requirements Traceability Matrix

Req ID	Requirement	Scope / Source	Relevance	Evidence	Test Case	Status / Result
REQ-PWR-009	Low battery warning appears despite full charge.	strong - RTM_M aster.xlsx	Matched complaint terms against power_manager; relevance score 9.	verified_current - Controlled current verification	TC-PWR-009	current and usable - Pass
REQ-PWR-001	Device does not power on.	strong - RTM_M aster.xlsx	Matched complaint terms against power_manager; relevance score 8.	verified_current - Controlled current verification	TC-PWR-001	current and usable - Pass
REQ-PWR-002	Device shuts down unexpectedly during measurement.	strong - RTM_M aster.xlsx	Matched complaint terms against power_manager; relevance score 8.	verified_current - Controlled current verification	TC-PWR-002	current and usable - Pass
REQ-PWR-006	Device overheats while charging.	strong - RTM_M aster.xlsx	Matched complaint terms against power_manager; relevance score 8.	verified_current - Controlled current verification	TC-PWR-006	current and usable - Pass
REQ-PWR-008	Device repeatedly restarts automatically.	strong - RTM_M aster.xlsx	Matched complaint terms against power_manager; relevance score 8.	verified_current - Controlled current verification	TC-PWR-008	current and usable - Pass
REQ-PWR-003	Battery drains unusually fast.	strong - RTM_M aster.xlsx	Matched complaint terms against power_manager; relevance score 7.	verified_current - Controlled current verification	TC-PWR-003	current and usable - Pass
REQ-PWR-004	Battery charging indicator does not appear.	strong - RTM_M aster.xlsx	Matched complaint terms against power_manager; relevance score 7.	verified_current - Controlled current verification	TC-PWR-004	current and usable - Pass
REQ-PWR-005	Battery does not charge fully.	strong - RTM_M aster.xlsx	Matched complaint terms against power_manager; relevance score 7.	verified_current - Controlled current verification	TC-PWR-005	current and usable - Pass
REQ-PWR-007	Power button is unresponsive.	strong - RTM_M aster.xlsx	Matched complaint terms against power_manager; relevance score 7.	verified_current - Controlled current verification	TC-PWR-007	current and usable - Pass
REQ-PWR-010	Incorrect battery percentage displayed.	strong - RTM_M aster.xlsx	Matched complaint terms against power_manager; relevance score 7.	verified_current - Controlled current verification	TC-PWR-010	current and usable - Pass
REQ-SPO 2-006	The device shall maintain operation on battery for the specified monitoring duration. [DEMO DATA - NOT CONTROLLED EVIDENCE]	medium - M1 requirement scope [DEMO DATA - NOT CONTROLLED EVIDENCE]	Matched complaint terms against Design Control; relevance score 4.	missing - Candidate evidence - requires Quality review [DEMO DATA - NOT CONTROLLED EVIDENCE]	TEST-006, TEST-026	usable with caution - Pass
REQ-ALR M-059	Alarm icon does not appear.	weak - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 2.	verified_current - Controlled current verification	TC-ALRM-059	current and usable - Pass

Trace Decay Risk Statement

Rows marked STALE are not filler rows: they are generated from the live Trace Decay result and indicate verification links that require review before the device can claim current-firmware evidence.

6. Design History File Index

#	Element	Document ID	Generated Unique Content	Status
1	Planning	DDP-DP-53B06935	Design plan for complaint-impact review	Open
2	Design Input	URS-DP-53B06935	Complaint-linked requirement review for The low battery icon appears even though the device continues to operate for a long time	Draft
3	Design Output	SDD-DP-53B06935	Power and Battery, Connectivity and Data Transfer design output references mapped	Draft
4	Design Review	DRM-DP-53B06935	Cross-functional review required	Pending
5	Verification	VPR-DP-53B06935	Objective evidence required for Power and Battery, Connectivity and Data Transfer on v3.4	Open
6	Validation	VAL-DP-53B06935	Use-condition validation required for The low battery icon appears even though the device continues to operate for a long time	Open
7	Design Transfer	DTR-DP-53B06935	Release readiness blocked until evidence reconciled	Pending
8	Design Changes	DCR-DP-53B06935	Change/service impact assessment linked to Power and Battery, Connectivity and Data Transfer	Open
9	DHF Index	DHFI-DP-53B06935	DocPlus+ index for DocPlus+ Pulse Oximeter	Generated

Validation Gap Check

The validation section remains open because the complaint describes a user-reported condition involving Power and Battery, Connectivity and Data Transfer. DocPlus+ therefore does not mark the DHF complete until validation, verification, or justified non-reproducibility evidence is linked.

7. EU MDR Technical Documentation - Annex II / III

Annex Ref	Section	DocPlus+ Generated Content
Annex II Section 1	Device Description	UDI UDI-DI-08717648-184649 scoped to complaint involving Power and Battery, Connectivity and Data Transfer.
Annex II Section 3	Design and Manufacturing Information	Design/change impact for Power and Battery, Connectivity and Data Transfer is carried into design output review.
Annex II Section 4	GSPR Compliance	Reported condition maps to safety/performance review: The low battery icon appears even though the device continues to operate for a long time.
Annex II Section 5	Benefit-Risk Analysis	Medium risk complaint requires residual-risk review before closure.
Annex II Section 6.1	Verification and Validation	Objective verification evidence must be reconciled against the complaint scope.
Annex III Section 1.3	PMS / PSUR Input	Complaint is retained as a post-market signal for trend monitoring.
Annex III Section 1.5	Vigilance Reporting	Reportability remains open until human regulatory review confirms decision.

Team NB Structure Check

The package preserves a cohesive structure with section titles, metadata, and cross-references so that information is locatable by a reviewer rather than buried in a generic AI narrative.

Record Boundary

The report is generated only from live application state: M1 graph context, M2 agent outputs, M4 evidence retrieval, AuditShadow findings, and Trace Decay alerts. It does not certify regulatory closure until the signature block is completed by authorized reviewers.

8. Complaint Investigation Workflow and Execution Trace

8.1 Structured Complaint Intake Detail

Device ID	DEV-PULSE-OX
Affected Component	COMP-POWER-BATTERY (COMP-POWER-BATTERY)
Severity	Low
Timeline	not specified
Symptom Codes	SYMPTOM_POWER_FAILURE, SYMPTOM_LOW, SYMPTOM_BATTERY, SYMPTOM_ICON, SYMPTOM_APPEARS, SYMPTOM_EVEN
Match Terms	appears, battery, power
Similar Prior Complaints	5

Complaint Narrative

The low battery icon appears even though the device continues to operate for a long time.

Related Prior Complaints

Complaint ID	Similarity	Summary	Source
CMP-006	4	User reported confusion between perfusion indicator and battery icon. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CPL-100-009	4	Low battery warning appears despite full charge. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CMP-003	2	Battery indicator dropped rapidly during extended monitoring. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CMP-012	2	MAUDE-style adverse event record linked to pulse oximeter use. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CPL-100-003	2	Battery drains unusually fast. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]

8.2 Investigation Chronology

Time	Agent	Action	Status	Detail
2026-07-08T10:38:19.249130Z	langgraph_supervisor	route to complaint_intake	info	Recorded
2026-07-08T10:38:19.249130Z	complaint_intake	extract structured complaint	started	Recorded
2026-07-08T10:39:58.194463Z	complaint_intake	extract structured complaint	completed	Elapsed 98944.7 ms
2026-07-08T10:39:58.195173Z	langgraph_supervisor	handoff check: similar incidents preserved for root cause	info	Structured event metadata recorded
2026-07-08T10:39:58.244289Z	langgraph_supervisor	route to root_cause	info	Recorded
2026-07-08T10:39:58.244289Z	root_cause	generate complaint-specific failure hypotheses	started	Recorded
2026-07-08T10:40:09.157061Z	root_cause	generate complaint-specific failure hypotheses	completed	Elapsed 10913.5 ms
2026-07-08T10:40:09.256867Z	langgraph_supervisor	route to firmware_traceability_ripple_check	info	Recorded
2026-07-08T10:40:09.257891Z	firmware_traceability_ripple_check	run firmware traceability ripple checks	started	Recorded
2026-07-08T10:40:09.264144Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T10:40:09.265323Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T10:40:12.991036Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T10:40:12.991036Z	firmware_traceability_ripple_check	run firmware traceability ripple checks	completed	Elapsed 3734.9 ms
2026-07-08T10:40:13.067065Z	langgraph_supervisor	route to evidence	info	Recorded
2026-07-08T10:40:13.067065Z	evidence	retrieve and validate evidence	started	Recorded
2026-07-08T10:40:14.314934Z	evidence	component-scoped evidence retrieval	info	Structured event metadata recorded
2026-07-08T10:40:14.314934Z	evidence	retrieve and validate evidence	completed	Elapsed 1248.0 ms
2026-07-08T10:40:14.397631Z	langgraph_supervisor	route to risk	info	Recorded

Chronology Interpretation

The sequence confirms the complaint passed through intake, root-cause hypothesis generation, firmware traceability ripple checking, evidence retrieval, risk scoring, CAPA drafting, and final reasoning. This preserves a reviewer-readable process trail for the AI investigation rather than presenting the CAPA as an unsupported narrative.

Trace AI - Pipeline Observability

Trace AI is operational engineering telemetry for this report generation. It records pipeline execution, LLM metadata, and retrieval/tool metadata only. It is not clinical evidence, not regulatory objective evidence, and not the same as Trace Decay or the Firmware Traceability Ripple Check.

Total wall-clock time	388772.88 ms
Agent steps	10
LLM calls	3
Tool/retrieval calls	25
Tokens	21887 total (14861 prompt / 7026 completion)
Longest agent	audit_shadow (133723.54 ms)

Agent Step Trace

Step ID	Agent	Action	Status	Elapsed	Sub-events
step-16e62b3fa2	complaint_intake	extract structured complaint	success	98944.8 ms	LLM 0; tools 4
step-f09cd2956d	audit_shadow	run adversarial audit	success	133723.54 ms	LLM 0; tools 5
step-67d4e062ba	cybersecurity	scan SBOM components against NVD	success	97454.86 ms	LLM 0; tools 3
step-8912c275be	root_cause	generate complaint-specific failure hypotheses	success	10913.51 ms	LLM 1; tools 2
step-01b8682e68	firmware_traceability_ripple_...	run firmware traceability ripple checks	success	3734.36 ms	LLM 0; tools 2
step-a73dbf0ad5	evidence	retrieve and validate evidence	success	1247.97 ms	LLM 0; tools 3
step-9ac50c7701	risk	calculate ISO 14971 risk	success	232.7 ms	LLM 0; tools 2
step-58d44fe790	capa	draft CAPA with citations	success	5705.77 ms	LLM 0; tools 4
step-c7b04c7039	openai_reasoning	final complaint reasoning	success	22373.43 ms	LLM 1; tools 0
step-10b379d9ad	openai_reasoning	final audit reasoning	success	14441.94 ms	LLM 1; tools 0

LLM Call Metadata

Call ID	Agent	Task	Model	Status	Tokens	Latency
llm-65c42ee683	root_cause	root_cause_hypotheses	gpt-4o-mini	success	5447 (4744/703)	10730.29 ms
llm-28f337ec8f	openai_reasoning	complaint_final_reasoning	gpt-5	success	8134 (4590/3544)	22370.97 ms
llm-926594c39c	openai_reasoning	audit_final_reasoning	gpt-5	success	8306 (5527/2779)	14436.42 ms

Tool and Retrieval Call Metadata

Call ID	Agent	Tool	Query / Target	Results	Latency
tool-39c685f7d7	complaint_intake	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		195.02 ms
tool-c72335deb4	audit_shadow	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		192.97 ms
tool-508ebaae13	cybersecurity	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		161.83 ms
tool-8a2148b230	complaint_intake	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		664.34 ms
tool-c5075db95b	audit_shadow	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		672.52 ms
tool-c7c3d7c22b	cybersecurity	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		725.09 ms
tool-77f7255da0	complaint_intake	graph_tools.get_device_context	device_id=DEV-PULSE-OX	15	93549.31 ms
tool-4a0a8a6e89	audit_shadow	graph_tools.get_device_context	device_id=DEV-PULSE-OX	15	94760.32 ms
tool-4f49d59649	audit_shadow	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		137.31 ms
tool-1ea30a9bcb	cybersecurity	graph_tools.get_device_context	device_id=DEV-PULSE-OX	15	97222.62 ms
tool-4aab99d330	complaint_intake	graph_tools.find_similar_complaints	term=The low battery icon appears even though the device continues to o...	5	90.9 ms
tool-fa56c32b1b	root_cause	graph_tools.get_graph_neighborhood	node_id=COMP-POWER-BATTERY	415	99.36 ms
tool-c0ec22b783	root_cause	graph_tools.get_graph_neighborhood	node_id=COMP-POWER-BATTERY	13	77.86 ms
tool-fa00d8b817	firmware_traceability_rip...	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		533.48 ms
tool-d11792f6a2	firmware_traceability_rip...	graph_tools.run_ripple	device_id=DEV-PULSE-OX; new_version=v3.5; changed_components=['COMP-POW...	10	3725.29 ms
tool-bf0da5fd0f	evidence	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		34.96 ms
tool-6db690552d	evidence	graph_tools.get_component_scoped_ev...	device_id=DEV-PULSE-OX; component_id=COMP-POWER-BATTERY; component_name...	12	753.55 ms
tool-1364dd707b	evidence	vector_tools.retrieve_documents_debug	device_id=DEV-PULSE-OX; query=The low battery icon appears even though...	0	493.03 ms
tool-5d0fb22ed3	risk	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		55.67 ms
tool-05b694403f	risk	graph_tools.open_capa_count	device_id=DEV-PULSE-OX	2	231.96 ms

5 additional tool/retrieval call(s) are retained in structured Trace AI data.

9. Root Cause Analysis

ID	Hypothesis	Probability	Affected Area	Key Why Chain
HYP-001	COMP-POWER-BATTERY: Stale Low-Voltage Alarm Overwriting Battery Full Flag [INFERRED DATA - REVIEW REQUIRED]	39%	COMP-POWER-BATTERY	Low battery icon appears on screen Device operates for extended periods despite indicator Possible stale alarm affecting status flags [INFERRED DATA - REVIEW REQUIRED]
HYP-002	COMP-POWER-BATTERY: Software/Firmware Bug Affecting Battery Status Display [DEMO DATA - NOT CONTROLLED EVIDENCE]	33%	COMP-POWER-BATTERY	Low battery icon is displayed incorrectly Device functions normally despite indicator No reset or power cycle resolving the issue [DEMO DATA - NOT CONTROLLED EVIDENCE]
HYP-003	COMP-POWER-BATTERY: Power Management Hardware Fault Leading to Incorrect Readings [DEMO DATA - NOT CONTROLLED EVIDENCE]	28%	COMP-POWER-BATTERY	Unexplainable low battery indication remains Device continues to work effectively without any apparent issue Potential discrepancy between hardware and software signaling [DEMO DATA - NOT CONTROLLED EVIDENCE]

HYP-001 Evidence Balance

Evidence For	CPL-100-009: Low battery warning appears despite full charge. [INFERRED DATA - REVIEW REQUIRED]
Evidence Against / Gaps	No evidence currently available confirming that a stale alarm condition is creating this issue. [INFERRED DATA - REVIEW REQUIRED]
Similar Incident Cross-Check	CPL-100-009 is consistent with this hypothesis, indicating that stale low-voltage alarms have been linked to incorrect battery status reporting. [INFERRED DATA - REVIEW REQUIRED]
Probability Rationale	The correlation with CPL-100-009 suggests a strong likelihood that a similar fault could be present, leading to a 70% probability. [INFERRED DATA - REVIEW REQUIRED]

HYP-002 Evidence Balance

Evidence For	Complaint report states the observed failure condition: The low battery icon appears even though the device continues to operate for a long time. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Evidence Against / Gaps	No direct evidence cites a software issue related to battery status indicators at this time. No evidence currently available confirming or ruling out any specific software or firmware bugs. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Similar Incident Cross-Check	Similar incidents do not directly implicate a software bug, but the nature of software/firmware issues often leads to similar display inaccuracies. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Probability Rationale	Given the nature of software issues in device operations, along with lack of contradiction, a 60% probability is assigned. [DEMO DATA - NOT CONTROLLED EVIDENCE]

HYP-003 Evidence Balance

Evidence For	Complaint report states the observed failure condition: The low battery icon appears even though the device continues to operate for a long time. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Evidence Against / Gaps	No complaints or evidence support a hardware malfunction relating to the battery management system. No evidence currently available confirming this hypothesis. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Similar Incident Cross-Check	CPL-100-003 involves battery performance issues, but does not confirm that the management hardware is faulty. Evidence is insufficient to determine a consistent pattern here. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Probability Rationale	The nature of electrical/mechanical faults cannot be discounted, but current data leads to only a 50% probability of this being the root cause. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Root Cause Rationale

1. HYP-001 (stale lowvoltage alarm overwriting full flag) is supported by similar incident CPL-100-009 (Closed - confirmed device failure; root cause: stale lowvoltage alarm persisted in nonvolatile settings). This matches the complaint symptom: low battery icon even when operation continues.
2. Controlled verification evidence shows related requirements/tests passed on v3.4: RUN-TC-PWR-009-V3-4 (linked to REQ-PWR-009), RUN-TC-PWR-002-V3-4 (REQ-PWR-002), and RUN-TC-PWR-003-V3-4 (REQ-PWR-003). This reduces the likelihood of recurrence on v3.4 but does not rule out version differences, edge conditions, or unit-specific persistence behavior.
3. HYP-002 (generic software/firmware display bug) remains possible given the nature of the symptom, but there is no direct evidence in the supplied records implicating a new software defect beyond the stale-alarm scenario already documented in CPL-100-009.
4. HYP-003 (power-management hardware fault) is less supported: no pattern of battery-management hardware failures is evidenced in the supplied incidents; CPL-100-003 concerned increased quiescent current from the optical AFE state, which explains fast drain but not a persistent low-battery icon with long runtime.
5. Human-factor/identification error is plausible per CMP-006 (confusion between perfusion indicator and battery icon), which could explain an apparent "low battery" indication while the device continues to function normally.

Primary Root Cause Position

For this generated draft, the leading position is that the complaint is associated with Power and Battery, Connectivity and Data Transfer and the evidence/control set tied to the reported condition. The position remains provisional until engineering review confirms or rules out the suspected cause with objective evidence.

10. Evidence and Provenance

Source	Evidence Class	Support	Conf.	Excerpt	Citation
REQ-PWR-009 / TC-PWR-009 / RUN-TC-PWR-009-V3-4	Controlled current verification	Yes	96%	REQ-PWR-009: Low battery warning appears despite full charge. Test TC-PWR-009: Verification test TC-PWR-009; result=Pass; firmware=v3.4. Evidence RUN-TC-PWR-009-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regress...	[Source: REQ-PWR-009/TC-PWR-009, Confidence: 0.96]
REQ-PWR-002 / TC-PWR-002 / RUN-TC-PWR-002-V3-4	Controlled current verification	Yes	96%	REQ-PWR-002: Device shuts down unexpectedly during measurement. Test TC-PWR-002: Verification test TC-PWR-002; result=Pass; firmware=v3.4. Evidence RUN-TC-PWR-002-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regre...	[Source: REQ-PWR-002/TC-PWR-002, Confidence: 0.96]
REQ-PWR-003 / TC-PWR-003 / RUN-TC-PWR-003-V3-4	Controlled current verification	Yes	96%	REQ-PWR-003: Battery drains unusually fast. Test TC-PWR-003: Verification test TC-PWR-003; result=Pass; firmware=v3.4. Evidence RUN-TC-PWR-003-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Verifi...	[Source: REQ-PWR-003/TC-PWR-003, Confidence: 0.96]
REQ-PWR-005 / TC-PWR-005 / RUN-TC-PWR-005-V3-4	Controlled current verification	Yes	96%	REQ-PWR-005: Battery does not charge fully. Test TC-PWR-005: Verification test TC-PWR-005; result=Pass; firmware=v3.4. Evidence RUN-TC-PWR-005-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Verifi...	[Source: REQ-PWR-005/TC-PWR-005, Confidence: 0.96]
REQ-PWR-007 / TC-PWR-007 / RUN-TC-PWR-007-V3-4	Controlled current verification	Yes	96%	REQ-PWR-007: Power button is unresponsive. Test TC-PWR-007: Verification test TC-PWR-007; result=Pass; firmware=v3.4. Evidence RUN-TC-PWR-007-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Verific...	[Source: REQ-PWR-007/TC-PWR-007, Confidence: 0.96]

Source	Evidence Class	Support	Conf.	Excerpt	Citation
REQ-PWR-008 / TC-PWR-008 / RUN-TC-PWR-008-V3-4	Controlled current verification	Yes	96%	REQ-PWR-008: Device repeatedly restarts automatically. Test TC-PWR-008: Verification test TC-PWR-008; result=Pass; firmware=v3.4. Evidence RUN-TC-PWR-008-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW...	[Source: REQ-PWR-008/TC-PWR-008, Confidence: 0.96]

10.1 Firmware Traceability Ripple Check

This section reports a firmware-traceability check: for each root-cause hypothesis, it examines whether linked verification test records match the complaint or candidate firmware version, or are stale/mismatched. This is a graph-based traceability signal, not a behavioral model or physical test of the device. It identifies where verification evidence may be out of date, not what the device would actually do under the reported failure condition.

Hypothesis	Analysis Type	Result	Interpretation	Evidence Class
HYP-001	Not applicable - alarm_timing_or_clinical_notification	not_run	HYP-001 evaluates COMP-POWER-BATTERY: Stale Low-Voltage Alarm Overwriting Battery Full Flag. The firmware traceability ripple check was not run because this failure mode maps to 'alarm_timing_or_clinical_notification', outside firmware-to-test traceability...	Firmware traceability signal - informational, not controlled verification evidence
HYP-002	Firmware traceability ripple check	fail	HYP-002 evaluates COMP-POWER-BATTERY: Software/Firmware Bug Affecting Battery Status Display. The firmware traceability ripple check found 10 linked verification test record(s) that do not match candidate firmware v3.5; this flags stale or mismatched verifi...	Firmware traceability signal - informational, not controlled verification evidence
HYP-003	Firmware traceability ripple check	fail	HYP-003 evaluates COMP-POWER-BATTERY: Power Management Hardware Fault Leading to Incorrect Readings. The firmware traceability ripple check found 10 linked verification test record(s) that do not match candidate firmware v3.5; this flags stale or mismatched...	Firmware traceability signal - informational, not controlled verification evidence

11. CAPA Report

Product	DocPlus+ Pulse Oximeter
Software Version	v3.4
CAPA Status	Pending Engineering Approval - Simulated Evidence Sufficient
Priority	Low
Risk Classification	Moderate - complaint requires corrective action review
Evidence Confidence	medium (score 0.91)
Evidence Class Mix	Controlled current verification: 10, Firmware traceability signal - informational, not controlled verification evidence: 2

Risk Evidence Confidence Basis

Weighted evidence-class score 0.91 from 12 item(s): Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=2. CAPA-blocking evidence-class item count: 2. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Engineering Approval Gate

This CAPA is supported by 10 controlled verification record(s); 2 simulated result(s) and 0 minor evidence-class gap(s) remain below the materiality threshold (0%). Formal Engineering review must approve the simulated/minor-gap findings and decide whether additional physical verification is required. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Problem Statement

A low priority complaint was opened for DocPlus+ Pulse Oximeter. The user reported: The low battery icon appears even though the device continues to operate for a long time. In practical terms, the concern is that the device may become unavailable or lose power during intended use. The complaint record confirms the reported condition; the exact technical cause and final correction remain under investigation until objective evidence is reviewed.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Complaint Summary

Reported condition: The low battery icon appears even though the device continues to operate for a long time. Affected area: Power and Battery, Connectivity and Data Transfer. Reported timing or trigger: Not fully specified by complaint text. Observed symptom signals: connectivity, pairing, or data-transfer failure. No patient or user injury is stated in the complaint text; the risk review must still confirm potential safety impact.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Immediate Containment Actions

Preserve the complaint evidence for Power and Battery, Connectivity and Data Transfer, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Investigation

Relevant device requirements, verification records, risk controls, and open quality actions were reviewed for the complaint scope. The investigation focused on the reported condition and the affected area: Power and Battery, Connectivity and Data Transfer. For this complaint the scoped area is Power and Battery, Connectivity and Data Transfer, with emphasis on power-path behavior, battery/charger characterization, service history, and safety-state evidence. Engineering and Quality reviewed the available design expectations, test records, risk controls, event evidence, and related quality actions. 2 verification records need Quality review; 2 related quality actions remain open. The scoped verification evidence is available for Quality review. Risk evidence basis: Weighted evidence-class score 0.91 from 12 item(s): Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=2. CAPA-blocking evidence-class item count: 2.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Root Cause Analysis

The current root-cause hypothesis is: The device continues to operate despite showing a low battery icon because a stale low-voltage alarm condition is overwriting the battery full status, leading to erroneous low battery indications. This is a probable cause, not a final confirmed cause, because the CAPA record still needs engineering reproduction results, configuration or software-version review, logs, and approved verification evidence showing whether the suspected cause is truly responsible.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Why CAPA Remains Open

The CAPA is not blocked for lack of evidence. The scoped evidence chain has controlled verification plus digital-twin or firmware-traceability simulation support, so the remaining gate is formal Engineering approval. Engineering must review the simulated findings against power availability and safe-state behavior, approve or reject their applicability, and record whether physical/controlled verification is still required. Risk evidence confidence is medium (score 0.91); Medium uncertainty: 10 of 12 evidence item(s) are current controlled verification, but 2 item(s) remain historical, simulated, candidate, or no-evidence class and cannot support CAPA closure. This CAPA is supported by 10 controlled verification record(s); 2 simulated result(s) and 0 minor evidence-class gap(s) remain below the materiality threshold (0%). It remains pending formal Engineering approval before closure; simulated results support the investigation but do not substitute for physical or controlled verification. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Risk Assessment

The main hazard is that the device may become unavailable or lose power during intended use. The current risk level is Medium and the complaint should remain in active risk review under ISO 13485 until patient impact, recurrence potential, and residual risk acceptability are documented. Risk evidence confidence is medium (score 0.91); Medium uncertainty: 10 of 12 evidence item(s) are current controlled verification, but 2 item(s) remain historical, simulated, candidate, or no-evidence class and cannot support CAPA closure.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Corrective Actions

Reproduce or simulate the reported condition for Power and Battery, Connectivity and Data Transfer under the reported use conditions. Software/System Engineering should review power-path behavior, battery/charger characterization, service history, and safety-state evidence, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Engineering must review and approve the simulated findings before closure; if Engineering rejects the simulation or deems it insufficient, open targeted physical or controlled verification before final approval.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Preventive Actions

Update the design, service, or release checklist so future changes affecting Power and Battery, Connectivity and Data Transfer include complaint-specific verification for connectivity, pairing, or data-transfer failure. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Verification of Effectiveness

Effectiveness is supported by controlled verification records plus simulated analysis of power availability and safe-state behavior. Before closure, an Engineering reviewer must approve the digital-twin or firmware-traceability simulation result, document why it is sufficient for this complaint, or require additional physical verification.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

CAPA Effectiveness Criteria

The CAPA is effective when the behavior for power availability and safe-state behavior is supported by current controlled records and the digital-twin or firmware-traceability simulation is formally approved by Engineering. Closure requires documented Engineering sign-off; additional physical verification is required only if Engineering does not accept the simulated support.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Lessons Learned

Future complaint handling for Power and Battery, Connectivity and Data Transfer should connect user-reported symptoms, objective evidence, risk controls, and verification records early. Similar changes or service decisions should include test coverage for the reported condition and review of whether existing controls remain effective.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

CAPA Closure

Closure decision: pending Engineering approval. This CAPA is supported by 10 controlled verification record(s); 2 simulated result(s) and 0 minor evidence-class gap(s) remain below the materiality threshold (0%). Formal Engineering review must approve the simulated/minor-gap findings and decide whether additional physical verification is required. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Technical evidence identifiers, requirement IDs, audit findings, and source citations are retained in the evidence and audit appendices. They are not repeated here so the CAPA remains readable for Quality, Engineering, Regulatory, and non-technical reviewers.

11.1 CAPA Action Plan

CAPA Record	CAPA-20260708-6935
Complaint Reference	CPL-20260708-53B0
CAPA Type	Corrective and preventive action
Current Status	Pending Engineering Approval - Simulated Evidence Sufficient
Engineering Approval Gate	This CAPA is supported by 10 controlled verification record(s); 2 simulated result(s) and 0 minor evidence-class gap(s) remain below the materiality threshold (0%). Formal Engineering review must approve the simulated/minor-gap findings and decide whether additional physical verification is required. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Action Plan

Purpose	Owner	Required Action	How Closure Is Proven
Immediate measures	Quality / Support	Preserve the complaint evidence for Power and Battery, Connectivity and Data Transfer, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact.	Evidence is preserved and simulated findings are available for Engineering review.
Fix the issue	Software/System Engineering	Reproduce or simulate the reported condition for Power and Battery, Connectivity and Data Transfer under the reported use conditions. Software/System Engineering should review power-path behavior, battery/charger characterization, service history, and safety-state evidence, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Engineering must review and approve the simulated findings before closure; if Engineering rejects the simulation or deems it insufficient, open targeted physical or controlled verification before final approval.	Engineering approves the digital-twin/simulated finding or records that physical verification is required.
Prevent recurrence	Engineering / Quality	Update the design, service, or release checklist so future changes affecting Power and Battery, Connectivity and Data Transfer include complaint-specific verification for connectivity, pairing, or data-transfer failure. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition.	Future changes include Engineering review of any simulated evidence before approval.
Check effectiveness	Quality Assurance	Effectiveness is supported by controlled verification records plus simulated analysis of power availability and safe-state behavior. Before closure, an Engineering reviewer must approve the digital-twin or firmware-traceability simulation result, document why it is sufficient for this complaint, or require additional physical verification.	Closure is proven by Engineering sign-off on the simulated support plus Quality/Regulatory approval.

Acceptance Criteria for Closure

Closure Gate	Required Evidence
Reported condition	The reported condition is supported by controlled records and simulated investigation findings.
Affected function	Power availability and safe-state behavior is covered by the scoped controlled and simulated evidence chain.
Engineering review	Engineering formally approves the digital-twin or firmware-traceability simulation, or requires physical verification.
Prototype disclaimer	Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.
Approval	Engineering, Quality, and Regulatory reviewers approve the completed CAPA record before closure.

12. Cybersecurity / SBOM Narrative

This section is device-level cybersecurity surveillance. It is not complaint evidence and it does not change the Risk Agent score automatically. It records what the Cybersecurity Agent found by reading the SBOM and querying the National Vulnerability Database for the CPE candidates in the workbook.

SBOM components	35
Components queried	25
NVD findings	7
Query date	2026-07-04
Source type	extracted
Cache status	cache_hit

Executive Severity Rollup

Critical	High	Medium	Low	None
0	2	5	0	0

VEX-style exploitability status is shown per CVE. The current automated SBOM/NVD match does not include device-specific exploitability determinations, so findings default to Under Investigation until Engineering documents whether the vulnerable code path is affected, fixed, not affected, or not reachable in this device configuration.

SBOM Component Coverage

Component	Version	Supplier	CPE Candidate	NVD Result
Bootloader	1.8.0	Device OEM / Secure Boot Team	cpe:2.3:o:device_oem:pulseox_bootloader:1.8.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Embedded Firmware	2.4.3	Device OEM / Embedded Software Team	cpe:2.3:o:device_oem:pulseox_firmware:2.4.3:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Operating System / RTOS	10.5.1	Amazon Web Services, Inc.	cpe:2.3:o:amazon:freertos:10.5.1:****:*	open_cves_found: 1 CVE record(s) returned by NVD for this CPE.
Device Drivers	1.6.2	Device OEM / Driver Integration Team	cpe:2.3:a:device_oem:pulseox_device_drivers:1.6.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Signal Processing Software	3.2.1	Device OEM / Clinical Algorithm Supplier	cpe:2.3:a:device_oem:pulseox_signal_processing:3.2.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Measurement Control Logic	2.1.0	Device OEM / Embedded Software Team	cpe:2.3:a:device_oem:pulseox_measurement_control:2.1.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Alarm Management Module	2.0.5	Device OEM / Safety Software Team	cpe:2.3:a:device_oem:pulseox_alarm_manager:2.0.5:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
User Interface Module	1.7.4	Device OEM / UI Software Team	cpe:2.3:a:device_oem:pulseox_ui_module:1.7.4:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Settings Configuration Manager	1.9.1	Device OEM / Embedded Software Team	cpe:2.3:a:device_oem:pulseox_settings_manager:1.9.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Data Storage Manager	1.5.0	Device OEM / Data Platform Team	cpe:2.3:a:device_oem:pulseox_data_storage_manager:1.5.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Patient Record Manager	1.3.2	Device OEM / Data Platform Team	cpe:2.3:a:device_oem:pulseox_patient_record_manager:1.3.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.

Component	Version	Supplier	CPE Candidate	NVD Result
Communication Stack	2.2.0	Device OEM / Connectivity Team	cpe:2.3:a:device_oem:pulseox_communication_stack:2.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Bluetooth Stack	5.3.0	Nordic Semiconductor ASA	cpe:2.3:a:nordicsemi:bluetooth_low_energy_stack:5.3.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Wi-Fi Stack	2.10	w1.fi project / Device OEM Integration	cpe:2.3:a:w1.fi:wpa_supplicant:2.10:****:	open_cves_found: 1 CVE record(s) returned by NVD for this CPE.
TCP/IP Stack	2.1.3	lwIP project	cpe:2.3:a:lwip_project:lwip:2.1.3:****:***	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
USB Interface Software	1.4.1	Device OEM / Connectivity Team	cpe:2.3:a:device_oem:pulseox_usb_interface:1.4.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
File System	3.8.1	littlefs project / Device OEM Integration	cpe:2.3:a:littlefs_project:littlefs:3.8.1:****:***	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Logging Module	1.2.8	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_logging_module:1.2.8:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Error Handling Module	1.1.6	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_error_handling:1.1.6:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Time Management Module	1.0.9	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_time_manager:1.0.9:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Self-Test Module	1.3.5	Device OEM / Test Software Team	cpe:2.3:a:device_oem:pulseox_self_test:1.3.5:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Firmware Update Manager	2.0.2	Device OEM / Release Engineering Team	cpe:2.3:a:device_oem:pulseox_update_manager:2.0.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Authentication Module	1.2.0	Device OEM / Security Software Team	cpe:2.3:a:device_oem:pulseox_authentication_module:1.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.

Component	Version	Supplier	CPE Candidate	NVD Result
Authorization Module	1.2.0	Device OEM / Security Software Team	cpe:2.3:a:device_oem:pulseox_authorization_module:1.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Cryptographic Library	3.0.13	OpenSSL Software Foundation	cpe:2.3:a:openssl:openssl:3.0.13:****:*	open_cves_found: 5 CVE record(s) returned by NVD for this CPE.

NVD CVE Findings - Structured Vulnerability Records

Vulnerability Record - CVE-2024-28115

CVE ID / NVD Reference	CVE-2024-28115 https://nvd.nist.gov/vuln/detail/CVE-2024-28115
CVSS Score / Severity / Vector	HIGH / CVSS 8.8 / v3.1 Vector: CVSS:3.1/ AV:L/ AC:L/ PR:L/ UI:N/ S:C/ C:H/ I:H/ A:H
CWE ID(s)	CWE-284: Improper Access Control, NVD-CWE-Other: Other weakness classification
Affected Component + Version	Operating System / RTOS 10.5.1
Vulnerability Description	NVD classifies the CVE under CWE-284, NVD-CWE-Other; review the public description for deployment-specific impact.
Exploitability / Exposure Context	Local or physically mediated exploitation should be assessed against service, update, or peripheral workflows. SBOM exposure: No direct exposure unless network stacks or external interfaces run as RTOS tasks.
Recommended Controls	review vendor fix status, confirm exploit preconditions, and document whether compensating controls cover the affected interface. SBOM controls: Minimal RTOS configuration, stack overflow checks, watchdog, task priority review, patch monitoring.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2024-28115

Vulnerability Record - CVE-2023-52160

CVE ID / NVD Reference	CVE-2023-52160 https://nvd.nist.gov/vuln/detail/CVE-2023-52160
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.5 / v3.1 Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:R/ S:U/ C:H/ I:N/ A:N
CWE ID(s)	CWE-287: Improper Authentication
Affected Component + Version	Wi-Fi Stack 2.10
Vulnerability Description	NVD describes an authentication or access-control bypass condition in the affected component.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Wi-Fi radio, WPA2/WPA3 negotiation, outbound cloud network.
Recommended Controls	update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: WPA2/WPA3, credential encryption, no ad-hoc mode, network allow-list, update monitoring.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2023-52160

Vulnerability Record - CVE-2009-1390

CVE ID / NVD Reference	CVE-2009-1390 https://nvd.nist.gov/vuln/detail/CVE-2009-1390
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-287: Improper Authentication
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes incomplete TLS certificate-chain validation, where a connection may be accepted because one certificate in an otherwise untrusted chain is trusted.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	verify full certificate-chain validation, require trusted root and hostname match, and regression-test rejected-chain cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-1390

Vulnerability Record - CVE-2009-3765

CVE ID / NVD Reference	CVE-2009-3765 https://nvd.nist.gov/vuln/detail/CVE-2009-3765
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-310: Cryptographic Issues
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3765

Vulnerability Record - CVE-2009-3766

CVE ID / NVD Reference	CVE-2009-3766 https://nvd.nist.gov/vuln/detail/CVE-2009-3766
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-310: Cryptographic Issues
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes a certificate or TLS validation weakness in the affected secure-communication path.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3766

Vulnerability Record - CVE-2009-3767

CVE ID / NVD Reference	CVE-2009-3767 https://nvd.nist.gov/vuln/detail/CVE-2009-3767
CVSS Score / Severity / Vector	MEDIUM / CVSS 4.3 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:N/ I:P/ A:N
CWE ID(s)	CWE-295: Improper Certificate Validation
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3767

Vulnerability Record - CVE-2019-0190

CVE ID / NVD Reference	CVE-2019-0190 https://nvd.nist.gov/vuln/detail/CVE-2019-0190
CVSS Score / Severity / Vector	HIGH / CVSS 7.5 / v3.1 Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:N/ S:U/ C:N/ I:N/ A:H
CWE ID(s)	NVD-CWE-noinfo: No CWE information
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes a denial-of-service condition that may let an attacker disrupt availability of the affected software path.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	patch affected code paths, constrain exposed interfaces, and confirm watchdog/recovery behavior for availability loss. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2019-0190

13. Electronic Records - 21 CFR Part 11

Requirement	DocPlus+ Implementation	Part 11
System Validation	DocPlus+ generation path is deterministic and testable through FastAPI route and PDF render checks.	11.10(a)
Audit Trail	Generation record links CPL-20260708-53B0, CAPA-20260708-6935, timestamp, and agent trace events.	11.10(e)
Access Controls	AI output is marked draft and requires QE/RA approval before controlled use.	11.10(d)
Electronic Signatures	Signature block captures reviewer, approver, date, and meaning of signature.	11.50
Record Integrity	PDF is generated as a stable artifact with a deterministic document number and retrievable filename.	11.10(c)
Human-Readable Copy	Selectable-text PDF is available through the document download endpoint.	11.10(b)
Retention	Artifact is retained under output/pdf and can be indexed by M5 or a QMS repository.	11.10(c)

Generation Log Summary

Generation Hash	53B06935
Input Fingerprint	53B06935D75175C8
AI Draft Status	Pending human quality and regulatory review
Low Confidence Handling	Evidence below 70% must be reviewed before release; current generated evidence remains visible with confidence values.

14. AuditShadow and Trace Decay

AuditShadow findings

Finding(s)	Requirement(s)	Risk	Observation	Remediation
FINDING-015, FINDING-016	REQ-PWR-002, REQ-PWR-005	Major	Requirement is touched by 1 open CAPA record(s), so readiness cannot be treated as clean.	Close or justify open CAPA impact before audit claim.
FINDING-024	REQ-SPO2-006	Major	M1 graph has no VERIFIED_BY test evidence linked for this requirement. M4 may have retrieved candidate evidence, but it is not controlled objective verification evidence until Quality links it to the requirement, test record, source, and version/configuration metadata. [DEMO DATA - NOT CONTROLLED EVIDENCE]	Link or create verification test evidence with current version/configuration metadata, then rerun AuditShadow. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Trace Decay alerts

Requirement(s)	Test(s)	Tested FW	Required FW	Reason
REQ-SPO2-006	TEST-006, TEST-026	v3.4	v3.4	Test is synthetic/demo backfill and cannot be treated as controlled verification evidence. [DEMO DATA - NOT CONTROLLED EVIDENCE]

14.1 Audit Data Interpretation and Corrective Action

AuditShadow and Trace Decay use the live M1 traceability graph as the controlled source for verification status. When the report flags missing evidence, it means the requirement does not have a formal VERIFIED_BY test link in that graph. It does not mean M4 found no supporting material. In this run, M4 retrieved 12 candidate evidence item(s), but candidate snippets remain draft support until a Quality reviewer links them to a requirement, test record, firmware version, and controlled source identifier.

Signal	What It Means	Source	Corrective Action
AuditShadow graph evidence gap	1 requirement finding(s) lack a VERIFIED_BY test link in M1.	M1 readiness score and traceability matrix	Create or link verification test evidence to each requirement and rescore readiness.
M4 retrieved evidence	12 candidate item(s) exist, but are not automatically controlled objective evidence.	M2 complaint pipeline and M4 evidence retrieval	Promote accepted snippets into controlled evidence nodes with requirement IDs, source, confidence, and firmware.
Trace Decay	2 firmware freshness alert(s) require review before release or approval.	M2 Trace Decay over M1 firmware/test links	Rerun verification on current firmware or document an approved equivalence rationale.
CAPA linkage	Open CAPA or complaint risk can keep readiness from being clean even when evidence exists.	M1 graph relationships and M2 risk assessment	Close, justify, or explicitly reference CAPA impact before final regulatory approval.

Reason and Corrective Action

The root issue is a traceability-linkage gap rather than a simple absence of all evidence. Corrective action is to synchronize M4 retrieved evidence back into the M1 graph as controlled verification evidence, then rerun AuditShadow and Trace Decay. Until that link exists, the report correctly treats the material as candidate support and keeps the audit finding open.

14.2 AuditShadow Detail Appendix

Finding(s)	Requirement(s)	Reference	Risk	Observation	Remediation
FINDING-015, FINDING-016	REQ-PWR-002, REQ-PWR-005	ISO 14971, IEC 62304	Major	Requirement is touched by 1 open CAPA record(s), so readiness cannot be treated as clean.	Close or justify open CAPA impact before audit claim.
FINDING-024	REQ-SPO2-006	IEC 60601-1	Major	M1 graph has no VERIFIED_BY test evidence linked for this requirement. M4 may have retrieved candidate evidence, but it is not controlled objective verification evidence until Quality links it to the requirement, test record, source, and version/configuration metadata. [DEMO DATA - NOT CONTROLLED EVIDENCE]	Link or create verification test evidence with current version/configuration metadata, then rerun AuditShadow. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Audit Interpretation

Major and critical findings should be triaged before report approval. A graph-linked evidence gap means the requirement is present in the M1 graph while still lacking a controlled verification test relationship acceptable for submission or inspection. Candidate RAG evidence must be reviewed, accepted, and linked before the finding can close.

14.3 Trace Decay Detail Appendix

Requirement(s)	Test(s)	Tested Firmware	Required Firmware	Reason
REQ-SPO2-006	TEST-006, TEST-026	v3.4	v3.4	Test is synthetic/demo backfill and cannot be treated as controlled verification evidence. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Traceability Impact Statement

Trace Decay identifies verification evidence that may no longer represent the active or proposed firmware state. These alerts are not final nonconformities by themselves, but they are mandatory review inputs for release, CAPA, and audit-readiness decisions.

15. Cross-Standard Rendering Matrix

Element	FDA / QMSR	EU MDR	ISO 13485
Safety Requirement	Design input under QMSR; linked to verification evidence.	GSPR and Annex I safety/performance claim.	ISO 13485 Clause 7.3.3 design input.
Test Evidence	DHF verification record; current configuration/version required.	Annex II Section 6.1 V&V evidence.	Clause 7.3.6 design verification.
Complaint Record	Complaint file CPL-20260708-53B0 with malfunction assessment.	Post-market vigilance signal under Articles 83/87.	Complaint handling under Clause 8.2.2.
CAPA Record	CAPA-20260708-6935 opened for Power and Battery, Connectivity and Data Transfer.	Trend/CAPA input for PMS and vigilance review.	Corrective action under Clause 8.5.2.
Risk Documentation	Risk analysis tied to design controls and complaint file.	Benefit-risk update in Annex II Section 5.	Risk management continuity through production/post-production.
Product Documentation	Lifecycle impact for Power and Battery, Connectivity and Data Transfer.	Design information and change-control evidence.	Validation and change-control evidence required.

Regulatory Lens Note

The same live graph and agent state are rendered under three regulatory lenses. This report therefore does not duplicate content blindly; it reformats the same evidence chain for FDA/QMSR, EU MDR, and ISO 13485 review contexts.

16. Document Control and Approval

Document No.	DOCPLUS-M4-20260708-53B06935
Revision	A
Revision Date	2026-07-08
Complaint ID	CPL-20260708-53B0
CAPA ID	CAPA-20260708-6935
Prepared by	DocPlus+ M4 Evidence and Documentation Layer
Generated from	Live M2 complaint pipeline, M1 graph context, M4 evidence retrieval, and device-level SBOM/NVD scan when attached
Generated at	2026-07-08T10:40:47Z
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Role	Name	Signature	Date
Quality Reviewer			
Regulatory Reviewer			